Unit: Unit 0: Foundations of Algebra
Topic: Isolating Variables

Name: ________________
Date: ___________

Foundation (Jalapeno)

Q1. Solve for x in $2x + 5 = 11$
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$

Q2. Rearrange the formula $P = 2l + 2w$ to isolate l
(A) $l = \frac{P-2w}{2}$
(B) $l = \frac{P+2w}{2}$
(C) $l = P - 2w$
(D) $l = P + 2w$
(E) $l = \frac{P}{2w}$

Q3. Solve for y in $y - 3 = 7$
(A) $y = 4$
(B) $y = 5$
(C) $y = 10$
(D) $y = 11$
(E) $y = 12$

Q4. Isolate x in $\frac{x}{4} = 9$
(A) $x = 36$
(B) $x = 40$
(C) $x = 45$
(D) $x = 35$
(E) $x = 32$

Q5. Rearrange $C = 2\pi r$ to solve for r
(A) $r = \frac{C}{2\pi}$
(B) $r = \frac{C}{\pi}$
(C) $r = 2\pi C$
(D) $r = \pi C$
(E) $r = C^2$

Q6. Solve for z in $z + 2 = 11$
(A) $z = 9$
(B) $z = 10$
(C) $z = 12$
(D) $z = 13$
(E) $z = 15$

Q7. Isolate t in $\frac{t}{3} = 12$
(A) $t = 30$
(B) $t = 36$
(C) $t = 40$
(D) $t = 45$
(E) $t = 35$

Q8. Rearrange the formula $A = \frac{1}{2}bh$ to isolate h
(A) $h = \frac{2A}{b}$
(B) $h = \frac{A}{2b}$
(C) $h = 2Ab$
(D) $h = A - 2b$
(E) $h = \frac{A}{b}$

Open Ended Questions (FRQ)

Q9. Solve the equation $4x + 2 = 3x + 5$ for x. Show all work.

Q10. The cost C of producing x units of a product is given by $C = 2000 + 5x$. Rearrange the equation to isolate x. Explain the context of your solution.

Chef's Tips

- Emphasize the importance of isolating variables in algebraic equations.
- Ensure students understand how to rearrange formulas to solve for specific variables.
- Note that proper notation and showing work are crucial for full credit.